

KARTHICK BALAJI T

Full Stack Developer | Backend Developer | AI Fullstack Developer

karthickbalaji028@gmail.com • +91-7339087528 • Chennai, TN • Open to Relocation
linkedin.com/in/karthickbalaji001 • github.com/karthickbalaji011220

PROFESSIONAL SUMMARY

Full Stack Developer with 8+ months of internship experience building 10+ production-ready web applications using Python, Django, React.js, and Java Spring Boot. Certified in AI Fullstack (A Grade, 87.13%) from Vetri Technology Solutions. Proficient in REST API design, JWT/OAuth2 authentication, MySQL, MongoDB, and deployment via GitHub and Render. Seeking Full Stack or Backend Developer roles in Chennai, Coimbatore, or Bangalore.

TECHNICAL SKILLS

Languages:	Python, Java, JavaScript (ES6+), TypeScript, SQL
Frontend:	React.js, HTML5, CSS3, Bootstrap 5, Responsive Design, REST API Integration
Backend:	Django, Django REST Framework (DRF), Spring Boot, Node.js, Flask
Databases:	MySQL, MongoDB, PostgreSQL, SQL Server, Django ORM, Query Optimisation
AI / Tools:	OpenAI GPT-4, Gemini API, Claude API, DeepSeek, TensorFlow, OpenCV, NLP, LLM Integration
DevOps:	Git, GitHub, Render, Docker, Postman, VS Code, Linux, CI/CD, Agile/Scrum

PROFESSIONAL EXPERIENCE

Vetri IT Systems Private Limited — AI Fullstack Developer Intern

Oct 2025 – Apr 2026

Vetri IT Systems Pvt. Ltd., Tenkasi, Tamil Nadu | Corporate ID: U62099TN2024PTC172387

- Built 10+ production-ready full-stack web applications using Python, Django, and React.js with JWT auth, REST APIs, and MySQL/MongoDB.
- Integrated OpenAI GPT-4, Gemini, Claude, and DeepSeek APIs for chatbots, text summarisation, and NLP dashboards across 3+ client projects.
- Engineered scalable RESTful APIs with Django REST Framework, reducing manual data processing by ~40%.
- Implemented JWT auth, OAuth2 flows, and Role-Based Access Control (RBAC) in Django and Spring Boot, securing 5+ applications.
- Developed mobile-first React.js dashboards with Bootstrap 5; worked in Agile/Scrum sprints with GitHub CI/CD pipelines.

Projectcadz Tech — Mobile Security Intern

Jul 2025 – Aug 2025

Projectcadz Tech, Coimbatore, Tamil Nadu

- Implemented AES encryption and data protection for mobile applications; identified 10+ vulnerabilities via threat analysis.
- Produced security audit documentation and testing procedures, streamlining compliance reviews across 2+ app modules.

PROJECTS

FoodHub — Full Stack Food Ordering Platform

2025

Tech Stack: Python, Django, MySQL, Razorpay, Google OAuth, Jazzmin, Bootstrap 5

- Built a Swiggy/Zomato-style food ordering system with live order tracking, cart management, and multi-restaurant dashboards.
- Integrated Razorpay payment gateway for secure online transactions and Google OAuth 2.0 for social login.
- Implemented custom Jazzmin admin panel for restaurant and menu management; resolved MariaDB 10.4 compatibility issues on XAMPP.
- Designed responsive UI with Bootstrap 5 and handled real-time order status updates using Django ORM and MySQL.

Billing Management System — Spring Boot

2025

Tech Stack: Java, Spring Boot, Thymeleaf, H2 Database, Chart.js, iText7, Spring Security

- Developed a full-featured billing system with role-based access control (admin/staff), invoice creation, and product/customer management.
- Built interactive Chart.js dashboards for sales analytics, revenue tracking, and monthly billing summaries.
- Implemented iText7 PDF invoice generation with custom formatting and automatic email dispatch via Gmail SMTP.
- Resolved complex Lombok annotation, BigDecimal precision, and H2 embedded database configuration issues during development.

MediCare HMS — Hospital Management System

Tech Stack: Python, Django, MariaDB, Razorpay, Gmail SMTP, ReportLab, Bootstrap 5

- Built an 11-module HMS covering patient registration, doctor scheduling, appointment booking, billing, and pharmacy management.
- Integrated Razorpay payment gateway for outpatient billing and Gmail SMTP for automated appointment confirmation emails.
- Generated detailed medical reports and patient discharge summaries as PDFs using ReportLab library.
- Designed role-based dashboards for admin, doctors, and receptionists with secure Django authentication and session management.

2025

Courier Tracking System — Spring Boot

Tech Stack: Java, Spring Boot, JWT, MySQL, Thymeleaf, Maven, Spring Security

- Delivered a 40-file Maven project with three role-based dashboards: admin, delivery personnel, and customer portals.
- Implemented JWT-based stateless authentication and Spring Security for secured API endpoints across all roles.
- Built real-time shipment status tracking with parcel assignment workflows, delivery confirmation, and history logs.
- Designed RESTful APIs for courier creation, route assignment, and status updates with MySQL persistence.

2025

Adaptive Visual Recommender — Dynamic Image Clustering

Tech Stack: Python, Django, OpenCV, SQL Server, K-Means, REST API

- Built a real-time K-Means image recommendation engine processing visual metadata to deliver personalised content suggestions.
- Boosted simulated user engagement by 35% and reduced integration turnaround time by 30% through API optimisation.
- Designed RESTful API endpoints for metadata management and image clustering with concurrent SQL Server query optimisation.
- Presented findings at ICA5NT 2024 and ICIT-24 international conferences on AI and network technologies.

Dec 2022 – Jul 2024

Change Detection in Remote Sensing Images using CNN

Tech Stack: Python, TensorFlow, NumPy, OpenCV, Keras

- Trained a deep CNN model on multi-temporal satellite imagery for binary change detection in remote sensing data.
- Applied data augmentation techniques and systematic hyperparameter tuning across 5+ experiment cycles to improve generalisation.
- Evaluated model performance with precision, recall, F1-score, and accuracy metrics on held-out unseen test datasets.
- Presented research outcomes at RISC 2024 international conference; findings contributed to published conference proceedings.

Nov 2021 – Apr 2022

EDUCATION

M.E. Computer Science Engineering

Hindusthan College of Engineering and Technology, Coimbatore • CGPA: 7.97

2024

B.E. Computer Science Engineering

Hindusthan College of Engineering and Technology, Coimbatore • CGPA: 7.71

2022

CERTIFICATIONS & AWARDS

- Certificate in AI Fullstack — Vetri Technology Solutions

A Grade (87.13%) | Cert No: CN3FTGZ7KN | Sep 2025 – May 2026 | ISO 9001:2015 Certified
- AI Fullstack Developer Internship Certificate — Vetri IT Systems Pvt. Ltd.

Oct 2025 – Apr 2026
- Meta Front-End Developer Professional Certificate — Coursera

Aug – Oct 2025
- Python Developer Bootcamp — Udemy

Mar – Aug 2022
- Research Papers — ICA5NT 2024, RISC 2024, ICIT-24

Presented 3 papers at International Conferences on AI & Network Technologies